

Real-Time Fraud Detection

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Validation

15 automated tests

Abstract

This project documents a fraud-scoring system built around the real constraints of authorization-time decisioning: rare positives, temporal behavior, and extremely low latency.

Project Background

Fraud scoring is not only a modeling problem. It is a streaming state-management problem where the useful signal comes from recent user and merchant behavior, and where train-serve skew can erase offline gains.

Headline Result

PR-AUC 0.974 (188x baseline); serving e2e p99 1.19 ms

Scope Overview

Synthetic transaction stream with bursty fraud behavior. Bounded-memory online feature store shared by train and serve paths. Leak-free offline training table and calibrated scorer. Serving benchmarks, dashboards, and validation tests.

Project Context

Streaming feature engineering + sub-millisecond online serving for card fraud - imbalanced, latency-critical, leakage-free.

Executive Summary

Background, objectives, scope, and project impact.

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Background

Fraud scoring is not only a modeling problem. It is a streaming state-management problem where the useful signal comes from recent user and merchant behavior, and where train-serve skew can erase offline gains.

Objectives

- Engineer time-aware fraud features without future leakage.
- Keep online feature serving within a sub-millisecond budget.
- Train and calibrate a model on an imbalanced event stream.

Scope

- Synthetic transaction stream with bursty fraud behavior.
- Bounded-memory online feature store shared by train and serve paths.
- Leak-free offline training table and calibrated scorer.
- Serving benchmarks, dashboards, and validation tests.

Impact

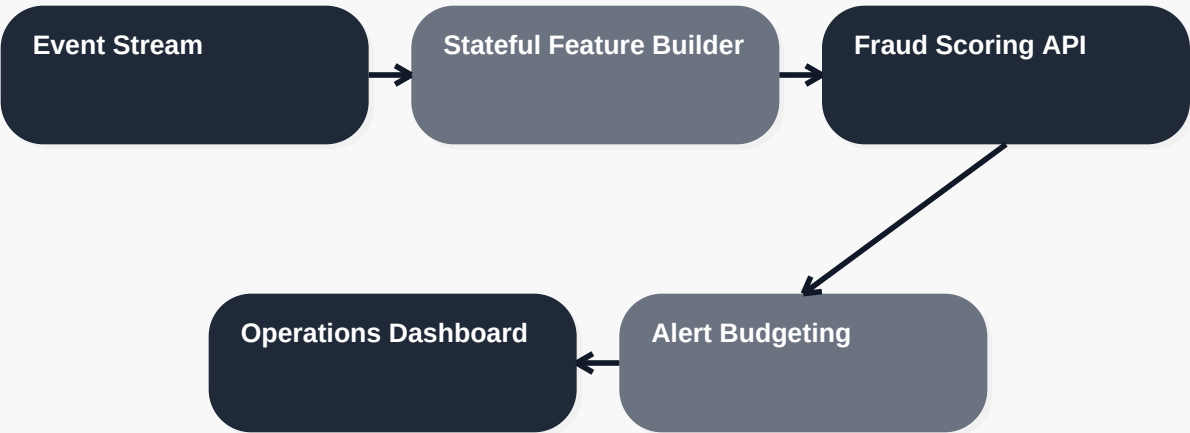
The finished system shows that disciplined feature engineering and leakage control matter more than model novelty. Strong precision-recall results were paired with production-like p99 latency instead of being traded off against it.

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Solution Architecture

The design optimizes for low-latency scoring with rolling behavioral features and a review budget that maps to real operations.

Core system flow



Design Note 1

The online feature store is the product core because that is where most fraud signal lives.

Design Note 2

Leakage prevention is enforced structurally through time ordering, not only by convention.

Design Note 3

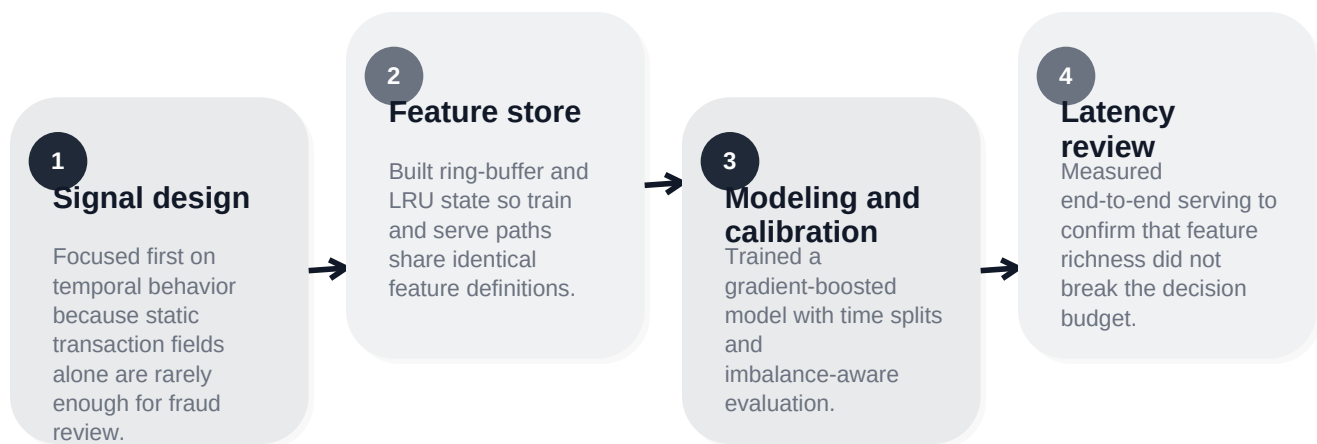
The system values operational latency and reviewability alongside PR-AUC gains.

Implementation Process

Delivery phases, engineering approach, and quality checkpoints.

Delivery Narrative

Streaming feature engineering + sub-millisecond online serving for card fraud - imbalanced, latency-critical, leakage-free.



Quality Checkpoints

- 15 automated tests validate core behavior and edge cases.
- Benchmark files and screenshots are generated from committed project artifacts.
- The run path stays deterministic so numbers can be reproduced and reviewed directly.

Engineering Principles

- The online feature store is the product core because that is where most fraud signal lives.
- Leakage prevention is enforced structurally through time ordering, not only by convention.
- The system values operational latency and reviewability alongside PR-AUC gains.

Evidence And Results

Test evidence, benchmark interpretation, and screenshot review.

15 passing tests

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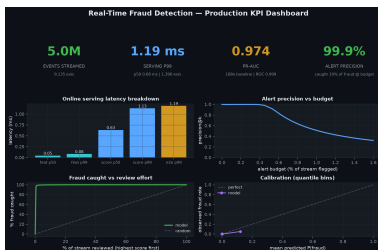
PR-AUC 0.974 (188x baseline); serving e2e p99
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Benchmark Evidence

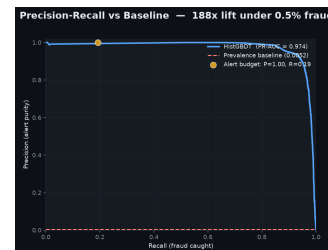
This repository emphasizes visual benchmark artifacts and automated validation outputs. Where tables are not present, the screenshots and test fixtures remain the primary review surface.

Result Interpretation

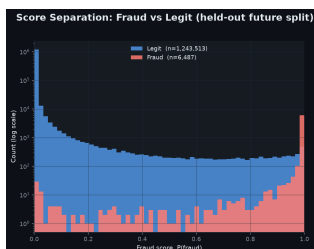
- PR-AUC gains are meaningful because the baseline is an imbalanced fraud task rather than a balanced toy dataset.
- The latency screenshots show the design stays practical for authorization-time scoring.
- Shared train and serve logic lowers the risk that offline gains disappear in



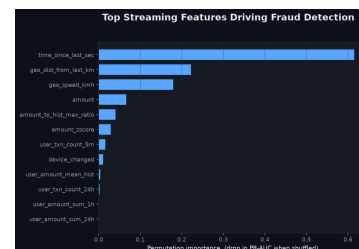
Production KPI dashboard



Precision-Recall vs baseline



Score distribution - fraud vs legit



Permutation feature importance

Risks And Next Steps

Operational considerations, mitigation choices, and forward path.

Risk Register

Train-serve skew Reduced by using the exact same feature computation path for offline and online modes.

Alert overload Managed with review-budget thinking and probability calibration.

Behavior drift Addressed by keeping the feature store stateful and by leaving room for retraining triggers.

Next Steps

- Add cost-sensitive threshold tuning for different fraud-loss profiles.
- Introduce drift monitoring on feature distributions and approval outcomes.
- Expand the review layer with case management and human feedback capture.

Closing Note

The project brief is intentionally written as delivered work rather than a transfer note. It documents the business problem, the engineering choices, the measured evidence, and the remaining operational path in a format suitable for portfolio review.

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